Fiscal Year:	FY 2013	**Task Last Updated:**	FY 11/13/2012
PI Name:	German, Randall M. Ph.D.		
Project Title:	Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	randgerman@gmail.com	**Fax:**	FY none
PI Organization Type:	UNIVERSITY	**Phone:**	858-922-4985
Organization Name:	San Diego State University		
PI Address 1:	College of Engineering		
PI Address 2:	282 Surfview Court		
PI Web Page:			
City:	Del Mar	**State:**	CA
Zip Code:	92014	**Congressional District:**	53
Comments:	co-investigator Eugene A Olevsky, SDSU, Elisa Torresani, SDSU, MS 1326		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	10/01/2010	**End Date:**	11/30/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	Pending NOTE: End date changed to 11/30/2013 (previously was 3/30/2014) per NSSC information (Ed., 1/28/14) NOTE: End date changed to 3/31/2014 per NSSC information (Ed., 6/18/13)		
Key Personnel Changes/Previous PI:	November 2012. Jose Alvarado joined as visiting professor; Aditya Bothate (MS) went to work, Timothy Young is his replacement; Amber Ritchey (BS) worked as undergraduate.		
COI Name (Institution):	Olevsky, Eugene (San Diego State University)		
Grant/Contract No.:	NNX10AV38G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A three year research project is proposed to investigate the gravity-dependent microstructure development and the related macroscopic shape evolution of multiphase powder materials during liquid phase sintering (LPS). The proposed research is aimed at achieving two critical goals: (i) in-depth analysis of the LPS induced pore-grain structure evolution by the de-convolution of the impact of gravity and (ii) exploring sintering under microgravity conditions as a technique for in-space fabrication and repair. Indeed, future NASA efforts to extend human exploration back to the Moon and beyond will require development of processes that permit fabrication under reduced gravity conditions. This capability is needed to reduce resource requirements for extraterrestrial construction and in space repair. Many proposed in-space fabrication and repair techniques involve particulate materials processing, such as via sintering, but limited data are available to assess the viability of these proposed net-shape techniques under reduced gravity. The research team will develop a computer code for the multi-scale simulation of LPS that will embrace both microstructure development and macroscopic shape distortion. Liquid phase sintering experiments on Earth and in microgravity will be conducted for a variety of heavy alloys to validate the sintering model framework. The anticipated research outcomes will be relevant to current and future space exploration needs for habitat creation, extraterrestrial exploration, and vehicle repair activities during trips to Mars or the Moon.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research is focused on liquid phase sintering, one of the most widely used manufacturing routes for ceramics, cemented carbides, powder metals, and composites. In the USA this is probably evaluated at $20 billion in annual product for metal cutting, oil well drilling, automotive engines, bearings, computers, and even watches. These components are designed for high performance and high precision, and are often fabricated in large production quantities. The net-shape fabrication, where final size and shape is achieved without machining, is a challenge since often dimensions change 15 to 20% due to shrinkage in going from the powder to the dense component. Thus, computer models of the sintering process are vital to proper tool and process design. Further, to attain specific mechanical, thermal, electrical or other properties requires further attention to microstructure control. This research provides a computational model that predicts final size and shape and with multiple scale model predicts properties. Further, with robust inclusion of gravitational gradients, liquid-solid migration, and substrate friction, it will be useful for predicting component attributes in a range of gravitational conditions as required for eventual fabrication on the moon, planets, or microgravity. For example, in building structure or repairing vehicles.
Task Progress:	Liquid phase sintering experiments are being conducted on a variety of tungsten heavy alloy compositions. Early experiments relied on tungsten-nickel-copper formulations. By adjusting the composition it is possible to change important parameters in liquid phase sintering: • more tungsten increases the density (the control parameter is the percentage of tungsten content where tungsten is 19.3 g/cc) • more nickel increases the solubility of the solid in the liquid and the wetting of the liquid on the solid and reduces the dihedral angle, while more copper has the opposite effects (the control parameter is the Ni to Cu ratio) • higher sintering temperature increases the coarsening rate (the control parameter is the peak temperature) After fully developing the W-Ni-Cu system, a temperature limitation was imposed by EAS, requiring a new lower temperature variant – W-Mn-Ni-Cu. This change proved difficult since manganese is very sensitive to oxygen (oxygen is released into the liquid phase by tungsten during liquid phase sintering). The scientific intent is to map a broad range of compositions for densification, distortion, and microstructure development and to couple both ground-based and microgravity experiments with robust models. Morphological transformations associated with liquid phase sintering would be measured with the two gravity variants, to document differences expected in the evolution of surface area, grain size, and density. The alloys have been identified, largely focused on tungsten heavy alloys with a high density. A modified alloy was needed to induce liquid formation at temperatures below 1200 C to remain compatible with the ESA materials science rack furnace. The use of a tungsten heavy alloy provides enhanced gravity roles in small samples so densification and distortion can be compared in similar samples under two gravitational conditions. Essentially the study is fully operation for ground-based studies. Sample preparation for eventual space experiments has started, and in the summer of 2012 work is progressing on vacuum quartz encapsulation. This will allow cartridge storage for up to two years without concern over leakage. The work on computer modeling has progressed using data from prior studies and the assumption of viscosity versus density is giving good predictions of density, distortion, and final microstructure. The framework for modeling is relying on mostly finite element analysis and should be compatible with standard engineering analysis. It is anticipated that Monte Carlo simulations will be employed to add microstructure development to the three dimensional component flow during sintering. The focus here is on grain growth and the linkage previously identified in porosity and grain size during densification. In a microgravity environment the pores will lose buoyancy and tend to undergo coarsening, leading to less densification and even bloating. Thus far, no model is able to explain these differences and this will hinder eventual fabrication and repair in low gravity environments. Meanwhile, existing data are being fit to a new porosity-grain size relation that details the contiguity of the solid structure to calculate the grain growth rate parameter. Current efforts are adding contiguity to this relation. All of this is in light of the newly developed W-Mn-Ni-Cu alloy custom fabricated for liquid phase sintering at 1200 C as required by ESA. The finite element calculations will attempt to include liquid flow and spreading into the viscous flow concepts to explain differences in densification and distortion during sintering. Presentations Since Award (Two Years): “Coarsening Models Applicable to Sintering,” invited keynote presentation, 2011 Annual Meeting, The Minerals, Metals, and Materials Society, San Diego, California, 1 March 2011. “Predictions of Tungsten Heavy Alloy Coarsening during Sintering,” contributed presentation, International Conference on Tungsten, Refractory Metals, and Hardmaterials VIII, San Francisco, California, 20 May 2011 (with E. A. Olevsky). “Modeling and Simulation for Sintering Process,” invited presentation, International Conference on Sintering 2011, Jeju, Korea, 29 August 2011 (with S. J. Park, S. H. Chung, Y. S. Kwon, S. T. Chung, S. G. Kim).

	"Coarsening during Sintering," contributed presentation, International Conference on Sintering 2011, Jeju, Korea, 31 August 2011. "Liquid Phase Sintering," invited TUBIK seminar, College of Engineering, TOBB Economic and Technological University, Ankara, Turkey, 4 October 2011. "Dynamic Model for Coarsening during Sintering," 2011 Materials Science and Technology Conference, Columbus Convention Center, Columbus, Ohio, 20 October 2011. "Multi-Scale Modeling and Experimentation on Liquid Phase Sintering Affected by Gravity: Preliminary Analysis," Materials Research in Microgravity Symposium, invited presentation, 141st Annual Meeting of the Metallurgical Society, Orlando, Florida, 14 March 2012 (with E. Olevsky, T. Young). "Sintering and Sintering Models," invited seminar, Adhesives Technology Division, Henkel Corp., Irvine, California, 5 April 2012. "Densification and Distortion of Tungsten Alloys Using Low Sintering Temperatures," presented at PowderMet 2012, MPIF-APMI, Nashville, Tennessee, 11 June 2012 (with W. Li, E. Olevsky, T. Young, J. McKittrick, A. Ritchey).
Bibliography Type:	Description: (Last Updated: 03/27/2025)
Abstracts for Journals and Proceedings	Bose A. Sadangi R, German RM. "A Review of Alloying in Tungsten Heavy Alloys." 141st Annual Meeting, The Minerals, Metals and Materials Society, Orlando, FL, March 11-15, 2012. TMS 2012. 141st Annual Meeting, The Minerals, Metals and Materials Society, Orlando, FL, March 11-15, 2012. p. 54. http://www.tms.org/Meetings/Annual-12/PDFs/TMS2012FinalProgramWEBwithAbstracts.pdf , Mar-2012
Articles in Peer-reviewed Journals	Fikes J, Park SJ, German RM. "Equilibrium states of liquid, solid, and vapor configurations for copper, tungsten, and pores in liquid phase Ssintering." Metallurgical and Materials Transactions B. 2011 Feb;42(1):202-9. http://dx.doi.org/10.1007/s11663-010-9460-7 , Feb-2011
Books/Book Chapters	German RM. "Thermodynamics of sintering." in "Sintering of advanced materials." Ed. Z.Z. Fang. Cambridge, UK : Woodhead Publishing Limited, 2010. p. 3-32. http://dx.doi.org/10.1533/9781845699949.1.3 , Sep-2010
Papers from Meeting Proceedings	German RM, Olevsky EA. "Predictions of Tungsten Alloy Coarsening During Sintering." Eighth International Conference on Tungsten, Refractory & Hardmaterials, San Francisco, CA, May 18–21, 2011. In: Tungsten, Refractory & Hardmaterials VIII. Proceedings of the Eighth International Conference on Tungsten, Refractory & Hardmaterials. Princeton, NJ : Metal Powder Industries Federation, 2011. p. 6.9-6.23. ISBN: 978-0-9819496-7-3. , Aug-2011
Papers from Meeting Proceedings	Bose A. Sadangi R, German RM. "A Review of Alloying in Tungsten Heavy Alloys." 141st Annual Meeting, The Minerals, Metals and Materials Society, Orlando, FL, March 11-15, 2012. In: TMS 2012 141st Annual Meeting and Exhibition, Supplemental Proceedings, Volume 1, Materials Processing and Interfaces. Hoboken, N.J. : Wiley, 2012. p. 455-465. ISBN: 978-1-1182-9607-3. , Apr-2012